

16S Metabarcoding with DADA2

A step-by-step tutorial using cucumber fermentation data

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Introduction

About this training

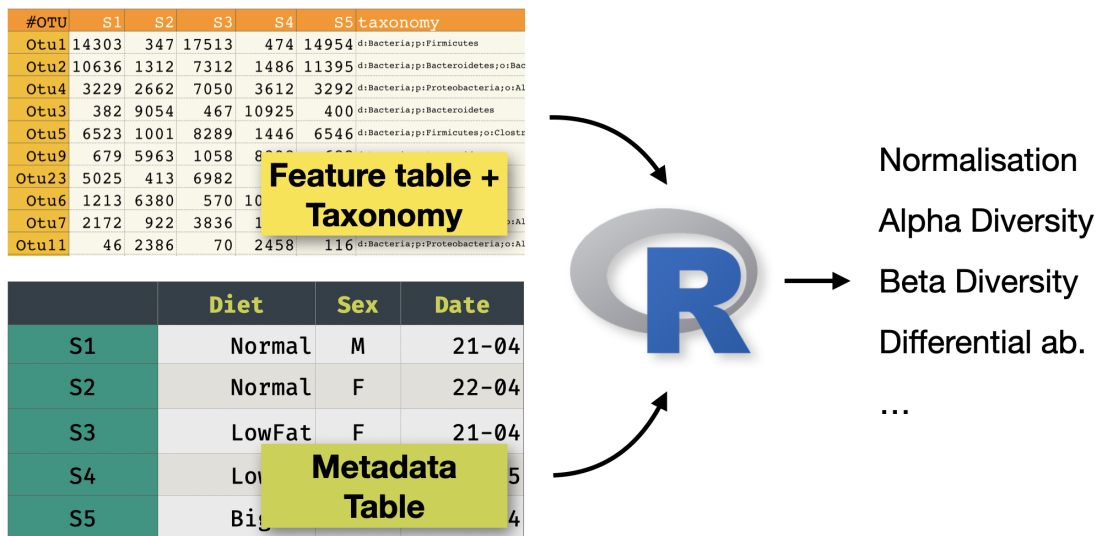


Figure 1: overview

This is a tutorial to analyse a metabarcoding dataset (in our case 16S) using DADA2. The input is a set of FASTQ files (typically, and in our case Illumina paired-end), and the output will be:

- A set of *representative sequences* (ASVs)
- A *feature counts table* (raw counts of ASVs per sample)
- The *taxonomy* of each representative sequence

We will use [DADA2](#), and for our example will use a recent dataset sequenced on NextSeq 2000, that uses a *binned quality* model that would not work nicely with the standard DADA2 tutorials.

Citation: Callahan *et al.* (2026) [DADA2: High-resolution sample inference from Illumina amplicon data](#)

Trainers

This session is designed and delivered by [QIB Core Bioinformatics](#) and in particular Andrea Telatin, Judit Talas and Alise Ponsero.

What is 16S metabarcoding?

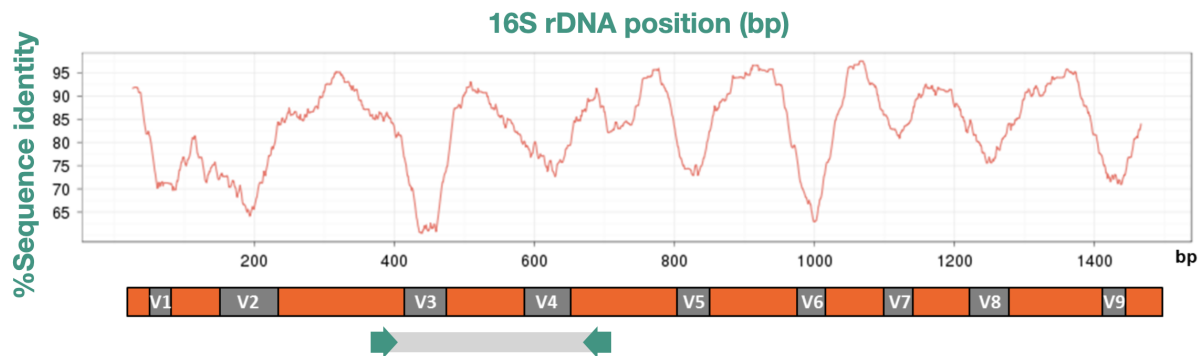


Figure 2: rDNA

16S metabarcoding is a culture-independent method to profile the bacterial communities present in a sample. It works by amplifying and sequencing a specific region of the bacterial 16S ribosomal RNA gene, a highly conserved gene found in virtually all bacteria.

A key feature of the gene is an alternation of very conserved regions (where we can design “universal primers” and hypervariable regions).

The typical workflow (*and its biases*) is:

1. Extract DNA from the sample (*are all cells breaking?*)
2. PCR-amplify the 16S gene using universal primers (*are we preferentially amplifying some taxa?*)
3. Sequence the amplicons on an Illumina platform (paired-end) (*we need to sequence only a fraction of the 1500bp gene!*)
4. Process the raw reads computationally to identify and quantify microbial taxa

ASVs vs OTUs

Historically, 16S data were analysed by clustering similar sequences into **Operational Taxonomic Units (OTUs)** at a fixed similarity threshold (typically 97%). One of the many popular packages offering OTU analysis is [Mothur](#). This approach is fast but loses resolution, meaning that sequences that differ by even one nucleotide are collapsed together. This approach was simultaneously dealing with sequencing errors and simplifying the identification of key taxa.

Modern pipelines instead resolve **Amplicon Sequence Variants (ASVs)**: exact, denoised sequences inferred from the data.

ASVs offer some benefits:

- Higher resolution — single-nucleotide differences are preserved
- Reproducibility — the same ASV sequence means the same biological entity across studies and platforms
- No arbitrary threshold — no need to choose a similarity cutoff (historically OTUs were clustered at 97% for no good reason)

But also some disadvantages:

- Some species will be [artificially split](#) into multiple ASVs (there is variability in 16S opersons in the same species)
- Some algorithms like DADA2 are sensitive to sequencing quality

DADA2 is the leading ASV-based pipeline and is what we will use throughout this tutorial.

The dataset

This tutorial uses real 16S amplicon data from a cucumber [fermentation study](#). The [metadata file](#) allows to download the raw reads.

Three types of fermentation were compared at multiple time points:

Group	Description
Lactobacillus	Brine inoculated with a <i>Lactobacillus</i> starter culture
Spontaneous	Brine fermented without a starter culture
Commercial	A commercially produced fermented cucumber product
Controls	Negative controls (fermentation and extraction blanks)

Time points: 24 hours (H24), 48 hours (H48), 1 week (W1), 2 weeks (W2). Each condition has three replicates. In total there are **29 samples**.

Samples were sequenced on an **Illumina NextSeq 2000** — an important detail that affects how we model sequencing errors (explained in Chapter 4).

Pipeline overview

Raw reads (FASTQ)

Quality assessment `plotQualityProfile()`

Filtering & trimming `filterAndTrim()`

Error rate modelling `learnErrors()` key step for NextSeq 2000

Denoising `dada()`

Merge paired reads `mergePairs()`

Sequence table `makeSequenceTable()`

Chimera removal `removeBimeraDenovo()`

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Taxonomy assignment `assignTaxonomy()`

Phyloseq object `phyloseq()`

Export (RDS, CSV)

Software requirements

All packages can be installed from [Bioconductor](#) / CRAN:

```
# Run once to install Bioconductor "installer"
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

# We can now install DADA2 and other packages
BiocManager::install(c("dada2", "phyloseq", "Biostrings"))

# And tidyverse...
install.packages(c("tidyverse"))
```

The analysis was developed and tested with:

- R 4.3
- dada2 1.28
- phyloseq 1.44
- tidyverse 2.0

1 How to Run This Tutorial

Before diving into the analysis, you need two things: the raw sequencing reads and a working R environment with the required packages. This preliminary page covers both.

1.1 TLDR; Minimal downloads

- A subset of the reads from zenodo.org/records/20082786
- Silva138.2 Database for DADA2 from zenodo.org/records/14169026
- A script to install the required packages from [GitHub](#)

1.2 The dataset: Getting the raw reads

The dataset used throughout this tutorial is publicly available in the **European Nucleotide Archive (ENA)** under accession [PRJEB112055](https://ena.ebi.ac.uk/ena/browser/view/PRJEB112055).

1.2.1 Downloading via the ENA browser

1. Go to <https://www.ebi.ac.uk/ena/browser/view/PRJEB112055>
2. Click the “**Download**” icon and select “**FASTQ files (FTP)**” to get a TSV file listing the FTP URLs for every sample.
3. Use `wget` to download the files listed in the TSV:

```
# Download the FTP links file from ENA
wget -i <(awk -F'\t' 'NR>1 {split($7,a,""); for(i in a) print a[i]}' filereport.tsv)
```

1.3 Running locally with RStudio

1.3.1 Install R and RStudio

1. Download and install **R** (4.3, but latest is preferred) from <https://cran.r-project.org/>
2. Download and install **RStudio Desktop** (free) from <https://posit.co/download/rstudio-desktop/>

1.3.2 Install required packages

Open RStudio and run the following once to install all dependencies:

```
# Install some tidyverse packages from CRAN
install.packages(c("tidyr", "tibble", "readr", "ggplot2", "patchwork"))

# Install the Bioconductor package manager
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

# Install DADA2 and phyloseq from Bioconductor
BiocManager::install(c("dada2", "phyloseq", "Biostrings"))
```

Installation can take several minutes the first time. Once complete, verify everything loaded correctly:

```
library(dada2)
```

Version check

The tutorial was developed with `dada2` 1.28, `phyloseq` 1.44, and `tidyverse` 2.0. If you see unexpected errors, check `packageVersion("dada2")` and update via `BiocManager::install("dada2")` if needed.

1.4 Running on the NBI HPC

HPC account required

You need an active NBI HPC account to follow this route. If you do not have one, contact your line manager or the NBI IT helpdesk to request access.

The NBI HPC provides RStudio through the **Open on Demand** web portal — no command-line setup required.

1.4.1 Step-by-step

1. Open your browser and go to <https://ood.hpc.nbi.ac.uk/>
2. Log in with your NBI HPC credentials.
3. In the top navigation bar click “**Interactive Apps**” and select **RStudio**.
4. Choose the compute resources you need (a 8 cores and 64 GB RAM are enough for this tutorial) and click “**Launch**”.
5. Wait for the job to start, then click “**Connect to RStudio Server**”.
6. Run the [installation script](#) to install dependencies

1.4.2 Installing packages on the HPC

The shared R installation on the HPC does **not** include DADA2, phyloseq, or tidyverse by default. Run the same installation commands as for a local setup — R will install the packages into your personal library (e.g. `~/r/lib/`):

```
if (!requireNamespace("BiocManager", quietly = TRUE))  
  install.packages("BiocManager")  
  
BiocManager::install(c("dada2", "phyloseq", "Biostrings"))  
  
install.packages("tidyverse")
```


2 Setup and Quality Assessment

2.1 Load libraries

We start by loading the packages needed for the entire analysis. It is good practice to load all dependencies at the top of the script so that any missing package is caught immediately.

```
library(dada2)      # Core DADA2 functions
library(tidyverse)  # Data manipulation and ggplot2
library(phyloseq)   # Microbiome data structures (used later)
```

2.2 Configure paths

We define all file paths relative to this project's working directory. Using relative paths makes the analysis portable — moving the project folder will not break the code.

```
# Directory containing the renamed, trimmed FASTQ files
# (reads are already primer-trimmed and renamed to clean sample names)
path_reads <- "../reads/"

# SILVA reference database for taxonomy assignment (used in Chapter 5)
path_silva <- "../db/silva_nr_v138_train_set.fa.gz"

# Directory for intermediate outputs
path_filtered <- "filtered"
path_precomp <- "precomputed"

dir.create(path_filtered, showWarnings = FALSE)
dir.create(path_precomp, showWarnings = FALSE)
```

2.3 Removing primers

DADA2 works on raw reads but after primer removal. We used [cutadapt](#) to remove the primers.

- Primer forward: CCTACGGGNGGCWGCAG
- Primer reverse: GGACTACHVGGGTATCTAATCC

The typical command to remove primers using cutadapt is:

```
cutadapt -a FWDPRIMER...RCREVPRIMER -A REVPRIMER...RCFWDPRIMER \
  --discard-untrimmed -m 100:100 -j 8 \
  -o sample1_noprimer_R1.fq.gz -p sample1_noprimer_R2.fq.gz \
  sample1_R1.fq.gz sample1_R2.fq.gz
```

- `-a` and `-A` provide the primers to trim, not the three dots `...` to separate the pairs. The first is applied to the first pair, while the second in the second pair. We provide *both* primers to each end for the case when the amplicon is shorter than the read length (e.g. ITS sequencing)
- `--discard-untrimmed` means not to save reads where the primers were not detected at all
- `-m X:Y` means not to save reads shorter than X (in R1) and Y (in R2), by default we could end up with empty reads
- `-o R1` and `-p R2` will specify the output files
- `-j INT` is for multithreading (in the example 8 cores used)
- the input are the positional arguments at the end (forward and reverse respectively)

2.4 Discover input files

DADA2 processes forward (R1) and reverse (R2) reads separately until the merging step. We find all FASTQ files matching each pattern and sort them to guarantee that the forward file at position `i` corresponds to the reverse file at position `i`.

```
fnFs <- sort(list.files(path_reads, pattern = "_R1.fastq.gz", full.names = TRUE))
fnRs <- sort(list.files(path_reads, pattern = "_R2.fastq.gz", full.names = TRUE))

cat("Forward read files found:", length(fnFs), "\n")
```

Forward read files found: 29

```
cat("Reverse read files found:", length(fnRs), "\n")
```

Reverse read files found: 29

2.4.1 Extract sample names

Sample names are extracted from the forward file names. Our files follow the naming convention `<SampleID>_R1.fastq.gz`, so we strip the `_R1.fastq.gz` suffix.

```
sample.names <- sub("_R1\\.fastq\\.gz$", "", basename(fnFs))
head(sample.names, 8)
```

```
[1] "Commercial_R1"      "Commercial_R2"      "Commercial_R3"
[4] "Control_extraction" "Control_fermentation" "Lactobacillus_H24_R1"
[7] "Lactobacillus_H24_R2" "Lactobacillus_H24_R3"
```

File naming matters

DADA2 assumes the forward and reverse file vectors are in the same order. Using `sort()` on both ensures correct pairing even when the filesystem returns files in an unexpected order.

2.5 Quality assessment

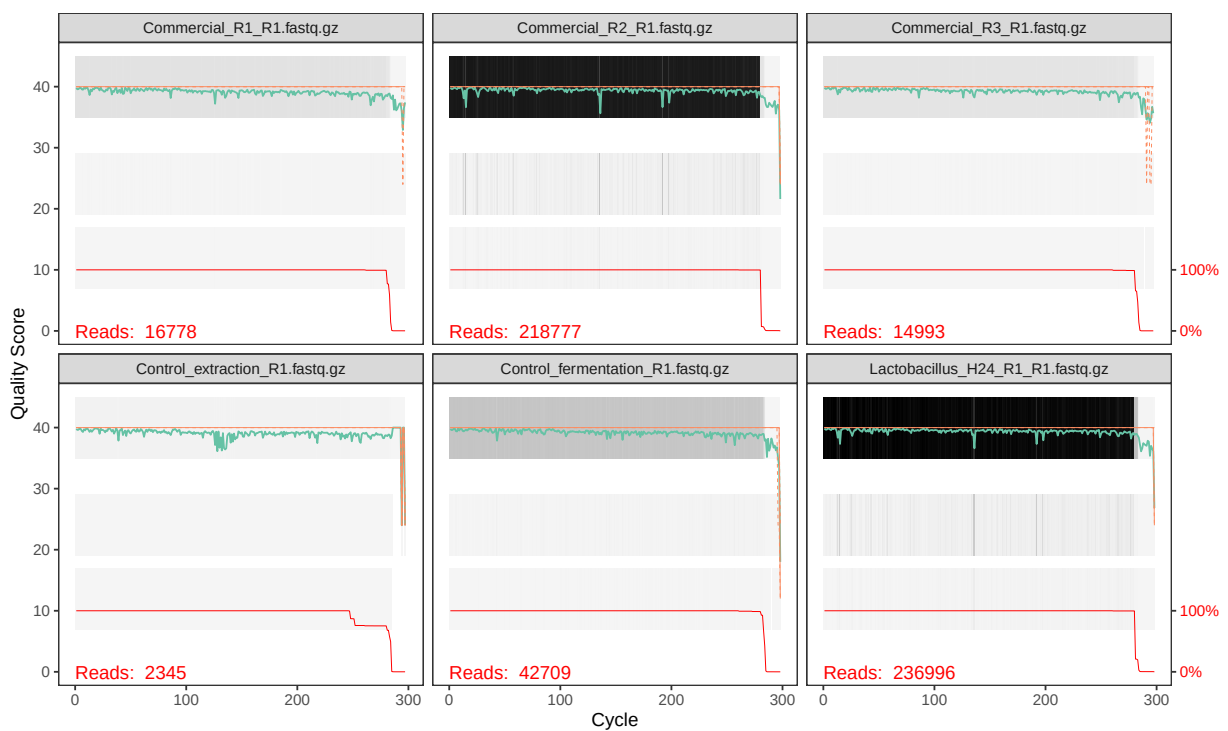
Before processing reads it is essential to visualise their quality profiles. This informs the truncation lengths we will use in the next chapter.

💡 Quick preview online

you can check the quality profile using our [utility website](#), without loading R at all.

2.5.1 Forward reads

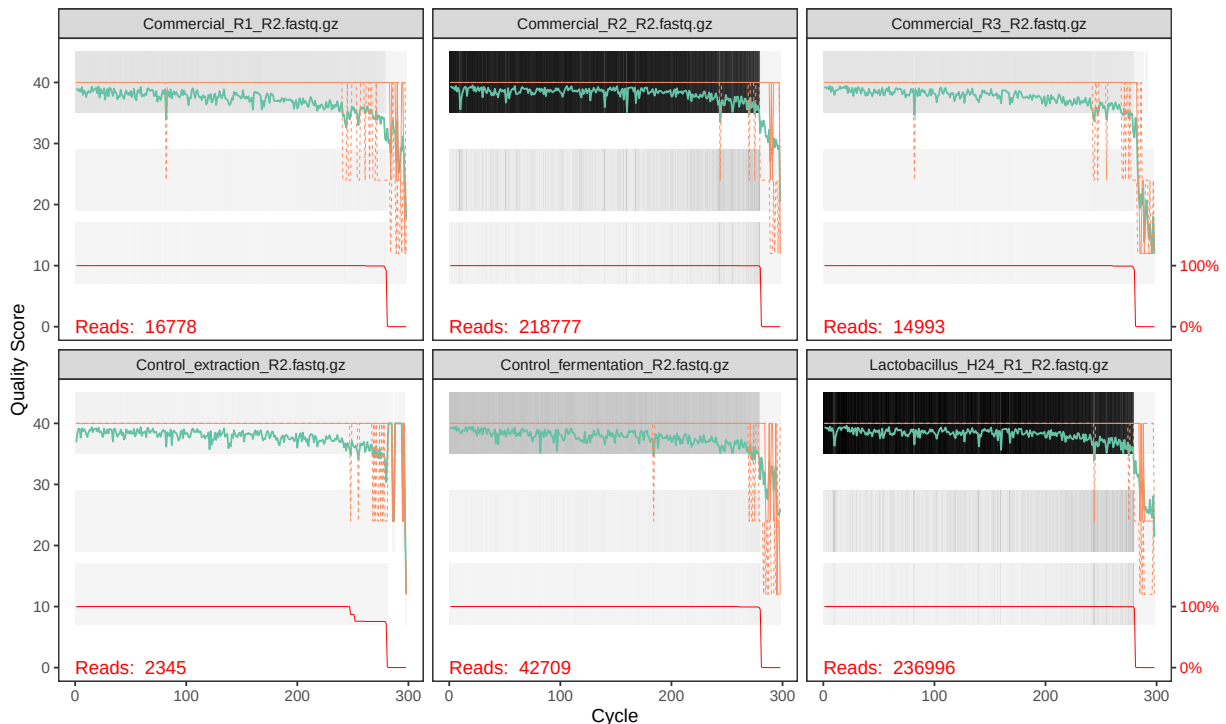
```
plotQualityProfile(fnFs[1:6])
```



2.5.2 Reverse reads

```
plotQualityProfile(fnRs[1:6])
```

2 Setup and Quality Assessment



2.6 Interpreting quality profiles

Each panel shows one sample. The *green line* is the mean quality score at each position, the *orange lines* are the quartiles, and the *red line* (at the bottom) shows the proportion of reads that extend to that position.

What to look for:

- **Drop in mean quality:** reads typically degrade towards the 3' end. Look for where the green line consistently falls below Q30.
- **Reverse reads are usually worse** than forward reads — this is normal for paired-end sequencing.
- **Truncation point:** you need to cut reads short enough to remove low-quality bases, but long enough so that the forward and reverse reads still overlap when merged.

Note that we also need to evaluate if we will be able to merge the two reads (if this is intended to happen). A typical V3-V4 amplicon is 460bp long, which means that 300bp paired end sequencing leaves a nice overlap to allow merging. If we need to trim a lot of bases so that their sum is going to be less than ~490bp, the overlap might not be sufficient.

2.6.1 Quality patterns in NextSeq 2000 data

This dataset was sequenced on an **Illumina NextSeq 2000**, which uses a 2-colour chemistry and reports **binned** (discrete) quality scores rather than the continuous Phred scores produced by older instruments like MiSeq. You will typically see quality scores clustering at a small number of levels (e.g. Q2, Q12, Q23, Q37) rather than spreading smoothly from 0 to 40.

This binning has an important consequence for DADA2's error modelling step — we will address it in Chapter 4.

i Choosing truncation lengths

Based on these profiles, we will truncate:

- **Forward reads** at position **260** (quality remains high to this point)
- **Reverse reads** at position **250** (slightly shorter to remove end degradation)

After truncation, the overlap between the two reads is sufficient for merging the V3–V4 amplicon (~460 bp expected product; reads of $260 + 250 = 510$ bp total gives ~50 bp overlap, which is adequate).

3 Filtering and Trimming

3.1 Why filter reads?

Raw sequencing reads contain:

- **Low-quality bases** at the 3' end that increase error rates in denoising
- **Ambiguous bases** (N) that DADA2 cannot handle
- **PhiX spike-in reads** — a control used for calibration during sequencing
- Occasionally, reads contaminated by the **PhiX bacteriophage genome**

The `filterAndTrim()` function removes all of these in a single step and also truncates reads to a fixed length, which is required for consistent error modelling.

3.2 Define filtered file paths

Here is where we will *save* the filtered reads.

```
filtFs <- file.path(path_filtered,
                    paste0(sample.names, "_F_filt.fastq.gz"))
filtRs <- file.path(path_filtered,
                    paste0(sample.names, "_R_filt.fastq.gz"))

# Give each vector element the corresponding sample name for easy indexing later
names(filtFs) <- sample.names
names(filtRs) <- sample.names
```

3.3 Run filterAndTrim

The *filterAndTrim* function allows to filter our reads that should be already free of sequencing primers!

```
out <- filterAndTrim(
  fnFs, filtFs,
  fnRs, filtRs,
  truncLen = c(260, 250), # truncate F reads at 260 bp, R reads at 250 bp
  maxN     = 0,           # discard any read with an ambiguous base
  maxEE    = c(2, 2),    # maximum expected errors per read (F and R separately)
  truncQ    = 10,        # truncate at first base with quality 10
  rm.phix   = TRUE,      # remove reads matching PhiX genome
  compress = TRUE,       # write gzip-compressed output
  multithread = TRUE     # use all available CPU cores
```

```
)
saveRDS(out, file.path(path_precomp, "out.rds"))
```

3.4 Parameter guide

Parameter	Value	Rationale
truncLen	c(260, 250)	Keeps high-quality region; must be long enough for reads to overlap after merging
maxN	0	DADA2 requires zero ambiguous bases
maxEE	c(2, 2)	Maximum <i>expected errors</i> (sum of error probabilities along the read); 2 is a standard threshold
truncQ	10	Removes the very-low-quality tail before the maxEE calculation
rm.phix	TRUE	Always advisable; PhiX contamination is common in Illumina data

What is Expected Errors (EE)?

The **expected errors** metric (introduced by Edgar & Flyvbjerg 2015) is more informative than a simple minimum quality threshold. For a read with quality scores $Q_1, Q_2, \dots, Q_n$, the EE is:

$$EE = \sum_{i=1}^n 10^{-Q_i/10}$$

A read with `maxEE = 2` can have no more than 2 expected sequencing errors in total. This is more lenient than requiring every base to exceed Q30, while still keeping read quality high.

3.5 Inspect filtering statistics

```
# Convert to data frame and add sample names
filter_stats <- as.data.frame(out)
filter_stats$sample <- rownames(filter_stats)
filter_stats$pct_kept <- round(100 * filter_stats$reads.out / filter_stats$reads.in, 1)

filter_stats[, c("sample", "reads.in", "reads.out", "pct_kept")]
```

	sample	reads.in
Commercial_R1_R1.fastq.gz	Commercial_R1_R1.fastq.gz	16778
Commercial_R2_R1.fastq.gz	Commercial_R2_R1.fastq.gz	218777

3.5 Inspect filtering statistics

Commercial_R3_R1.fastq.gz	Commercial_R3_R1.fastq.gz	14993
Control_extraction_R1.fastq.gz	Control_extraction_R1.fastq.gz	2345
Control_fermentation_R1.fastq.gz	Control_fermentation_R1.fastq.gz	42709
Lactobacillus_H24_R1_R1.fastq.gz	Lactobacillus_H24_R1_R1.fastq.gz	236996
Lactobacillus_H24_R2_R1.fastq.gz	Lactobacillus_H24_R2_R1.fastq.gz	271744
Lactobacillus_H24_R3_R1.fastq.gz	Lactobacillus_H24_R3_R1.fastq.gz	185402
Lactobacillus_H48_R1_R1.fastq.gz	Lactobacillus_H48_R1_R1.fastq.gz	232087
Lactobacillus_H48_R2_R1.fastq.gz	Lactobacillus_H48_R2_R1.fastq.gz	277388
Lactobacillus_H48_R3_R1.fastq.gz	Lactobacillus_H48_R3_R1.fastq.gz	257945
Lactobacillus_W1_R1_R1.fastq.gz	Lactobacillus_W1_R1_R1.fastq.gz	237242
Lactobacillus_W1_R2_R1.fastq.gz	Lactobacillus_W1_R2_R1.fastq.gz	283367
Lactobacillus_W1_R3_R1.fastq.gz	Lactobacillus_W1_R3_R1.fastq.gz	135637
Lactobacillus_W2_R1_R1.fastq.gz	Lactobacillus_W2_R1_R1.fastq.gz	270194
Lactobacillus_W2_R2_R1.fastq.gz	Lactobacillus_W2_R2_R1.fastq.gz	262219
Lactobacillus_W2_R3_R1.fastq.gz	Lactobacillus_W2_R3_R1.fastq.gz	280448
Spontaneous_H24_R1_R1.fastq.gz	Spontaneous_H24_R1_R1.fastq.gz	260410
Spontaneous_H24_R2_R1.fastq.gz	Spontaneous_H24_R2_R1.fastq.gz	297828
Spontaneous_H24_R3_R1.fastq.gz	Spontaneous_H24_R3_R1.fastq.gz	258467
Spontaneous_H48_R1_R1.fastq.gz	Spontaneous_H48_R1_R1.fastq.gz	311391
Spontaneous_H48_R2_R1.fastq.gz	Spontaneous_H48_R2_R1.fastq.gz	302367
Spontaneous_H48_R3_R1.fastq.gz	Spontaneous_H48_R3_R1.fastq.gz	295072
Spontaneous_W1_R1_R1.fastq.gz	Spontaneous_W1_R1_R1.fastq.gz	266958
Spontaneous_W1_R2_R1.fastq.gz	Spontaneous_W1_R2_R1.fastq.gz	271112
Spontaneous_W1_R3_R1.fastq.gz	Spontaneous_W1_R3_R1.fastq.gz	295075
Spontaneous_W2_R1_R1.fastq.gz	Spontaneous_W2_R1_R1.fastq.gz	291669
Spontaneous_W2_R2_R1.fastq.gz	Spontaneous_W2_R2_R1.fastq.gz	259494
Spontaneous_W2_R3_R1.fastq.gz	Spontaneous_W2_R3_R1.fastq.gz	314666

	reads.out	pct_kept
Commercial_R1_R1.fastq.gz	13226	78.8
Commercial_R2_R1.fastq.gz	178641	81.7
Commercial_R3_R1.fastq.gz	12109	80.8
Control_extraction_R1.fastq.gz	1419	60.5
Control_fermentation_R1.fastq.gz	34239	80.2
Lactobacillus_H24_R1_R1.fastq.gz	193289	81.6
Lactobacillus_H24_R2_R1.fastq.gz	221122	81.4
Lactobacillus_H24_R3_R1.fastq.gz	150467	81.2
Lactobacillus_H48_R1_R1.fastq.gz	189922	81.8
Lactobacillus_H48_R2_R1.fastq.gz	227640	82.1
Lactobacillus_H48_R3_R1.fastq.gz	210626	81.7
Lactobacillus_W1_R1_R1.fastq.gz	191040	80.5
Lactobacillus_W1_R2_R1.fastq.gz	233210	82.3
Lactobacillus_W1_R3_R1.fastq.gz	107509	79.3
Lactobacillus_W2_R1_R1.fastq.gz	219755	81.3
Lactobacillus_W2_R2_R1.fastq.gz	213873	81.6
Lactobacillus_W2_R3_R1.fastq.gz	230240	82.1
Spontaneous_H24_R1_R1.fastq.gz	209712	80.5
Spontaneous_H24_R2_R1.fastq.gz	241630	81.1
Spontaneous_H24_R3_R1.fastq.gz	209072	80.9
Spontaneous_H48_R1_R1.fastq.gz	253245	81.3
Spontaneous_H48_R2_R1.fastq.gz	243119	80.4
Spontaneous_H48_R3_R1.fastq.gz	238579	80.9

3 Filtering and Trimming

Spontaneous_W1_R1_R1.fastq.gz	217397	81.4
Spontaneous_W1_R2_R1.fastq.gz	218668	80.7
Spontaneous_W1_R3_R1.fastq.gz	239236	81.1
Spontaneous_W2_R1_R1.fastq.gz	237723	81.5
Spontaneous_W2_R2_R1.fastq.gz	208982	80.5
Spontaneous_W2_R3_R1.fastq.gz	256480	81.5

```
cat("Median reads retained:", median(filter_stats$pct_kept), "%\n")
```

Median reads retained: 81.2 %

```
cat("Minimum reads retained:", min(filter_stats$pct_kept), "%\n")
```

Minimum reads retained: 60.5 %

A retention rate above **70–80 %** is generally acceptable. If many samples lose more than 30 % of reads at this step, consider relaxing `maxEE` slightly (e.g. to 5) or revisiting the primer-trimming step upstream.

3.6 Check filtered files exist

A small sanity check: some samples may have lost all reads (e.g. negative controls with very low input). DADA2 will error if it tries to open an empty or absent file.

```
# Keep only samples for which filtered files were actually created
exists_F <- file.exists(filtFs)
exists_R <- file.exists(filtRs)

if (any(!exists_F | !exists_R)) {
  dropped <- sample.names[!exists_F | !exists_R]
  message("The following samples had no reads after filtering and will be excluded: ",
    paste(dropped, collapse = ", "))
  filtFs <- filtFs[exists_F & exists_R]
  filtRs <- filtRs[exists_F & exists_R]
  sample.names <- names(filtFs)
}

cat("Samples remaining:", length(sample.names), "\n")
```

Samples remaining: 29

4 Error Rate Modelling

4.1 What are error rates?

Every sequencing instrument makes mistakes. A base may be called as A when it was actually C. The probability of each such substitution depends on the quality score reported for that position: a base called at Q30 has a 1-in-1000 chance of being wrong, while a Q20 base has a 1-in-100 chance.

DADA2's denoising algorithm uses a **parametric error model** — learned from the data itself — to estimate the true frequency of each type of error at each quality score. This model is then used to distinguish genuine biological variants from PCR/sequencing errors.

4.2 The problem with binned quality scores

The standard DADA2 error estimation assumes that error rates vary continuously and monotonically with the reported quality score: higher quality $\rightarrow$ lower error rate. This assumption holds well for **MiSeq** data.

NextSeq 2000 and **NovaSeq** use a 2-colour chemistry that reports quality scores in a small number of discrete levels (typically Q2, Q12, Q23, Q37). This **binning** creates a mismatch: many bases with very different true error rates share the same quality label, and some quality levels may have zero or very few observations. The default `loessErrfun` fails to fit a smooth, monotone curve through such sparse, clustered data.

The solution is to use a modified loess fitting function that:

1. Adds a pseudocount to avoid $\log(0)$ issues
2. Adjusts the loess bandwidth and weighting
3. Enforces monotonicity explicitly after fitting

We provide four variants (Options 1–4). **Option 4** is recommended for NextSeq 2000 and NovaSeq data and is the one used in this tutorial.

4.3 The four error model variants

All four variants share the same overall structure. The differences are in how `loess()` is called:

Option	Span	Degree	Weights	Best for
1	2 (very wide)	default (2)	$\log_{10}(\text{tot})$	Heavily binned data
2	default	default (2)	tot (counts)	Standard MiSeq-like
3	default	default (2)	$\log_{10}(\text{tot})$	Intermediate
4	0.95	1	$\log_{10}(\text{tot})$	NextSeq / NovaSeq

Option 4 uses `degree = 1` (locally linear rather than locally quadratic) and a wide `span = 0.95`, which prevents over-fitting to the sparse quality bins and produces smooth, monotone error curves.

4.4 Define the error model functions

```
# Option 4: recommended for NextSeq 2000 / NovaSeq binned quality scores
loessErrfun_mod4 <- function(trans) {
  qq <- as.numeric(colnames(trans))
  est <- matrix(0, nrow = 0, ncol = length(qq))

  for (nti in c("A", "C", "G", "T")) {
    for (ntj in c("A", "C", "G", "T")) {
      if (nti != ntj) {
        errs <- trans[paste0(nti, "2", ntj), ]
        tot <- colSums(trans[paste0(nti, "2", c("A", "C", "G", "T")), ])
        rlogp <- log10((errs + 1) / tot) # +1 pseudocount
        rlogp[is.infinite(rlogp)] <- NA

        df <- data.frame(q = qq, errs = errs, tot = tot, rlogp = rlogp)

        # Locally-linear loess with log-count weights - the key change for binned data
        mod.lo <- loess(rlogp ~ q, df,
                        weights = log10(tot),
                        degree = 1,
                        span = 0.95)

        pred <- predict(mod.lo, qq)
        # Extend predictions to quality levels with no observations
        maxrli <- max(which(!is.na(pred)))
        minrli <- min(which(!is.na(pred)))
        pred[seq_along(pred) > maxrli] <- pred[[maxrli]]
        pred[seq_along(pred) < minrli] <- pred[[minrli]]
        est <- rbind(est, 10^pred)
      }
    }
  }

  # Clamp to a sensible range
  MAX_ERROR_RATE <- 0.25
  MIN_ERROR_RATE <- 1e-7
  est[est > MAX_ERROR_RATE] <- MAX_ERROR_RATE
  est[est < MIN_ERROR_RATE] <- MIN_ERROR_RATE

  # Enforce monotonicity: error rate at quality X must be error rate at quality 40
  estorig <- est
  est <- est %>%
    data.frame() %>%
    mutate(across(everything(), ~ if_else(. < X40, X40, .))) %>%
```



```

    as.matrix()
rownames(est) <- rownames(estorig)
colnames(est) <- colnames(estorig)

# Build the full 4x4 error matrix (including self-transitions)
err <- rbind(
  1 - colSums(est[1:3, ]), est[1:3, ],
  est[4, ], 1 - colSums(est[4:6, ]), est[5:6, ],
  est[7:8, ], 1 - colSums(est[7:9, ]), est[9, ],
  est[10:12, ], 1 - colSums(est[10:12, ])
)
rownames(err) <- paste0(rep(c("A", "C", "G", "T"), each = 4), "2",
                        c("A", "C", "G", "T"))
colnames(err) <- colnames(trans)
return(err)
}

```

i Options 1, 2, and 3 (click to expand)

These are provided for comparison. If you are processing MiSeq data you may find Option 2 (standard weights) or Option 3 (log-weight without degree/span change) more appropriate.

```

loessErrfun_mod1 <- function(trans) {
  qq <- as.numeric(colnames(trans))
  est <- matrix(0, nrow = 0, ncol = length(qq))
  for (nti in c("A","C","G","T")) {
    for (ntj in c("A","C","G","T")) {
      if (nti != ntj) {
        errs <- trans[paste0(nti,"2",ntj),]
        tot <- colSums(trans[paste0(nti,"2",c("A","C","G","T")),])
        rlogp <- log10((errs+1)/tot)
        rlogp[is.infinite(rlogp)] <- NA
        df <- data.frame(q=qq, errs=errs, tot=tot, rlogp=rlogp)
        mod.lo <- loess(rlogp ~ q, df, weights=log10(tot), span=2)
        pred <- predict(mod.lo, qq)
        maxrli <- max(which(!is.na(pred))); minrli <- min(which(!is.na(pred)))
        pred[seq_along(pred)>maxrli] <- pred[[maxrli]]
        pred[seq_along(pred)<minrli] <- pred[[minrli]]
        est <- rbind(est, 10^pred)
      }
    }
  }
  MAX_ERROR_RATE <- 0.25; MIN_ERROR_RATE <- 1e-7
  est[est>MAX_ERROR_RATE] <- MAX_ERROR_RATE; est[est<MIN_ERROR_RATE] <- MIN_ERROR_RATE
  estorig <- est
  est <- est %>% data.frame() %>%
    mutate(across(everything(), ~ if_else(. < X40, X40, .))) %>% as.matrix()
  rownames(est) <- rownames(estorig); colnames(est) <- colnames(estorig)
  err <- rbind(1-colSums(est[1:3,]), est[1:3,], est[4,], 1-colSums(est[4:6,]),
    est[5:6,], est[7:8,], 1-colSums(est[7:9,]), est[9,],
    est[10:12,], 1-colSums(est[10:12,]))
  rownames(err) <- paste0(rep(c("A","C","G","T"),each=4),"2",c("A","C","G","T"))
  colnames(err) <- colnames(trans)
  return(err)
}

loessErrfun_mod2 <- function(trans) {
  qq <- as.numeric(colnames(trans))
  est <- matrix(0, nrow = 0, ncol = length(qq))
  for (nti in c("A","C","G","T")) {
    for (ntj in c("A","C","G","T")) {
      if (nti != ntj) {
        errs <- trans[paste0(nti,"2",ntj),]
        tot <- colSums(trans[paste0(nti,"2",c("A","C","G","T")),])
        rlogp <- log10((errs+1)/tot)
        rlogp[is.infinite(rlogp)] <- NA
        df <- data.frame(q=qq, errs=errs, tot=tot, rlogp=rlogp)
        mod.lo <- loess(rlogp ~ q, df, weights=tot)
        pred <- predict(mod.lo, qq)
        maxrli <- max(which(!is.na(pred))); minrli <- min(which(!is.na(pred)))
        pred[seq_along(pred)>maxrli] <- pred[[maxrli]]
        pred[seq_along(pred)<minrli] <- pred[[minrli]]
        est <- rbind(est, 10^pred)
      }
    }
  }
  MAX_ERROR_RATE <- 0.25; MIN_ERROR_RATE <- 1e-7
  est[est>MAX_ERROR_RATE] <- MAX_ERROR_RATE; est[est<MIN_ERROR_RATE] <- MIN_ERROR_RATE

```

4.5 Learn error rates from the data

`learnErrors()` processes a subset of reads (here, up to 10^{10} bases, effectively all of them) to estimate per-cycle, per-substitution error rates. This is the most computationally demanding step of the pipeline.

The results are cached to disk so that re-rendering the document does not require re-running this step.

```
errF_file <- file.path(path_precomp, "errF.rds")
errR_file <- file.path(path_precomp, "errR.rds")

if (!file.exists(errF_file)) {
  message("Learning forward error model (this may take several minutes)...")
  errF <- learnErrors(filtFs, multithread = TRUE, nbases = 1e10,
                     errorEstimationFunction = loessErrfun_mod4, verbose = TRUE)
  saveRDS(errF, errF_file)
} else {
  message("Loading pre-computed forward error model.")
  errF <- readRDS(errF_file)
}

if (!file.exists(errR_file)) {
  message("Learning reverse error model (this may take several minutes)...")
  errR <- learnErrors(filtRs, multithread = TRUE, nbases = 1e10,
                     errorEstimationFunction = loessErrfun_mod4, verbose = TRUE)
  saveRDS(errR, errR_file)
} else {
  message("Loading pre-computed reverse error model.")
  errR <- readRDS(errR_file)
}
```

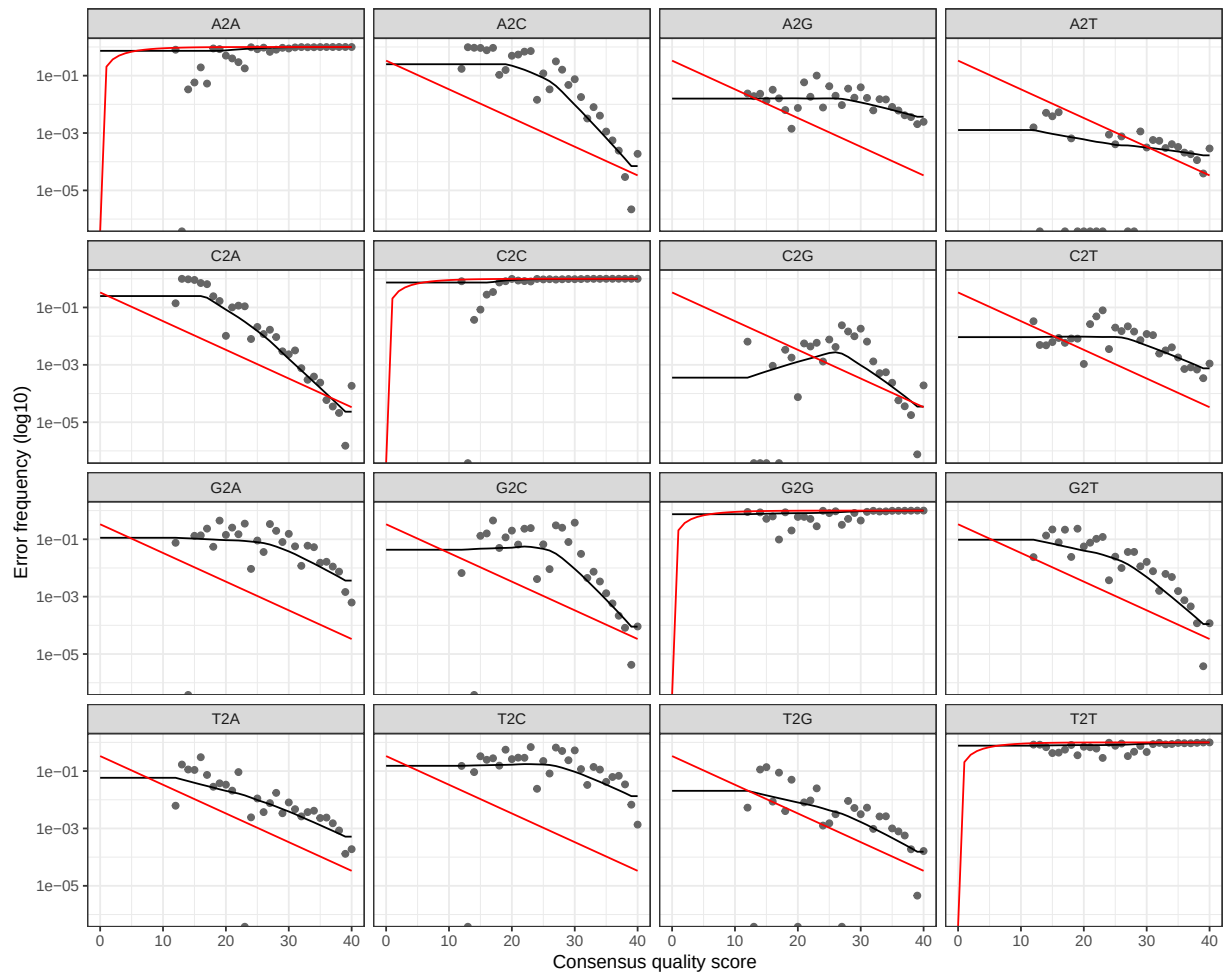
4.6 Visualise the error model

A good error model should show:

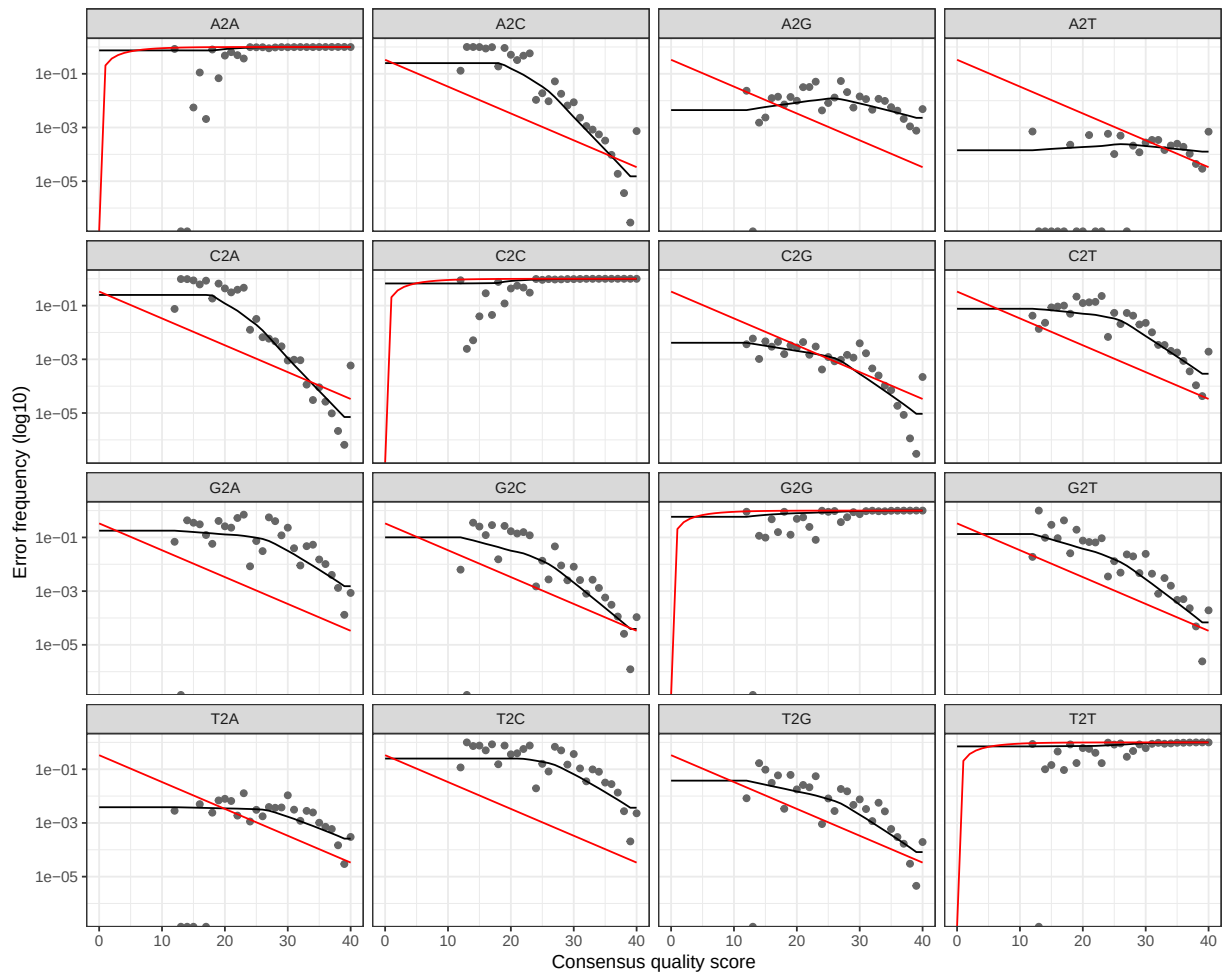
- **Black dots** (observed error rates) closely following the **black line** (fitted model)
- **Red line** (expected error rates based on the nominal quality score) close to the fitted line
- Error rates that **decrease monotonically** as quality increases

```
plotErrors(errF, nominalQ = TRUE)
```

4 Error Rate Modelling



```
plotErrors(errR, nominalQ = TRUE)
```



⚠ What a bad error model looks like

If the fitted line (black) diverges strongly from the observed points (dots), or if the curve is not monotone decreasing, the error model is unreliable. In that case, try a different option (1–3). Binned quality data often show clusters of points at a few quality levels with large gaps in between — Option 4 handles this best.

5 Denoising, Merging and Chimera Removal

5.1 Denoising with DADA2

The core DADA2 algorithm uses the error model learned in the previous chapter to distinguish true biological sequences from sequences that arose through PCR or sequencing errors. It works by asking: *given the error model, is this slightly different sequence more likely to be a genuine variant, or an error-derived copy of a more abundant sequence?*

The result is a set of **Amplicon Sequence Variants (ASVs)**: exact sequences with no artificial clustering.

```
dadaFs_file <- file.path(path_precomp, "dadaFs.rds")
dadaRs_file <- file.path(path_precomp, "dadaRs.rds")

if (!file.exists(dadaFs_file)) {
  message("Denoising forward reads...")
  dadaFs <- dada(filtFs, err = errF, multithread = TRUE)
  saveRDS(dadaFs, dadaFs_file)
} else {
  dadaFs <- readRDS(dadaFs_file)
}

if (!file.exists(dadaRs_file)) {
  message("Denoising reverse reads...")
  dadaRs <- dada(filtRs, err = errR, multithread = TRUE)
  saveRDS(dadaRs, dadaRs_file)
} else {
  dadaRs <- readRDS(dadaRs_file)
}
```

5.1.1 What does the dada2 output tell us?

```
# Inspect the first sample's denoising result
dadaFs[[1]]
```

```
dada-class: object describing DADA2 denoising results
90 sequence variants were inferred from 4843 input unique sequences.
Key parameters: OMEGA_A = 1e-40, OMEGA_C = 1e-40, BAND_SIZE = 16
```

The output reports how many unique sequences were identified in the sample and how many reads support each one. The ratio of reads to unique sequences gives an indication of how much error correction was performed.

5.2 Merging paired-end reads

After denoising, forward and reverse reads are merged to reconstruct the full amplicon sequence. Merging requires a minimum overlap region between the denoised forward and reverse reads — by default, at least 12 bp.

```
mergers_file <- file.path(path_precomp, "mergers.rds")

if (!file.exists(mergers_file)) {
  message("Merging paired reads...")
  mergers <- mergePairs(dadaFs, filtFs, dadaRs, filtRs, verbose = TRUE)
  saveRDS(mergers, mergers_file)
} else {
  mergers <- readRDS(mergers_file)
}
```

Merging failure

If a large proportion of reads fail to merge (visible in the tracking table later), the most common causes are:

1. **Insufficient overlap** — increase `truncLen` for one or both reads, or reduce it if the reads are too short
2. **Primer residues** — verify that primers were fully removed before this analysis
3. **Wrong amplicon size** — the reads must cover the amplicon with some overlap

5.3 Build the sequence table

`makeSequenceTable()` produces a sample $\times$ ASV matrix of read counts. This is the central data object in DADA2.

```
seqtab <- makeSequenceTable(mergers)
cat("Dimensions (samples  $\times$  ASVs):", dim(seqtab), "\n")
```

```
Dimensions (samples  $\times$  ASVs): 29 18934
```

5.3.1 Sequence length distribution

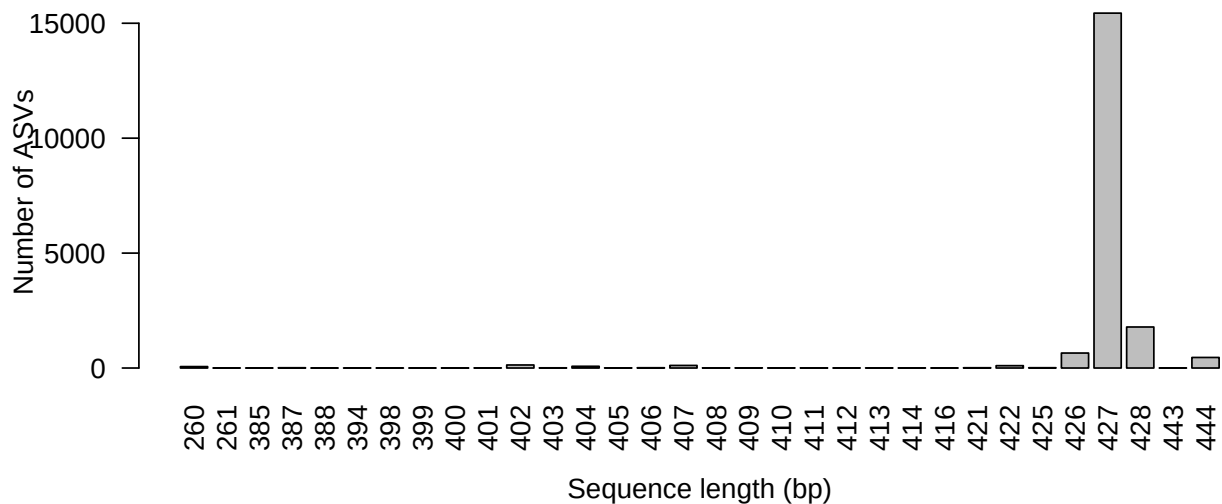
The expected amplicon length for the V3–V4 region is approximately 400–470 bp. Any sequences outside this range are likely artefacts (non-specific amplification or chimeras) and will often be removed in subsequent steps.

```
seq_lengths <- table(nchar(getSequences(seqtab)))
print(seq_lengths)
```


260	261	385	387	388	394	398	399	400	401	402	403	404
62	2	1	12	1	1	1	1	2	5	132	8	75
405	406	407	408	409	410	411	412	413	414	416	421	422
5	12	112	7	7	5	6	3	4	3	2	12	102
425	426	427	428	443	444							
13	652	15440	1785	2	459							

```
# Visualise
barplot(seq_lengths,
        xlab = "Sequence length (bp)",
        ylab = "Number of ASVs",
        main = "ASV length distribution (before chimera removal)",
        las = 2)
```

ASV length distribution (before chimera removal)



5.4 Chimera removal

Chimeric sequences are PCR artefacts created when an incomplete extension product from one cycle acts as a primer in a subsequent cycle, fusing the 5' end of one template with the 3' end of another. DADA2 identifies chimeras using the `removeBimeraDenovo()` function, which checks whether each sequence can be reconstructed from two more-abundant “parent” sequences.

```
seqtab_file <- file.path(path_precomp, "seqtab_nochim.rds")

if (!file.exists(seqtab_file)) {
  message("Removing chimeras...")
  seqtab.nochim <- removeBimeraDenovo(seqtab,
                                     method      = "consensus",
                                     multithread = TRUE,
                                     verbose     = TRUE)

  saveRDS(seqtab.nochim, seqtab_file)
} else {
  seqtab.nochim <- readRDS(seqtab_file)
```

```
}

cat("ASVs before chimera removal:", ncol(seqtab), "\n")
```

ASVs before chimera removal: 18934

```
cat("ASVs after chimera removal:", ncol(seqtab.nochim), "\n")
```

ASVs after chimera removal: 1091

```
pct_retained <- round(100 * sum(seqtab.nochim) / sum(seqtab), 1)
cat("Proportion of reads retained after chimera removal:", pct_retained, "%\n")
```

Proportion of reads retained after chimera removal: 89.8 %

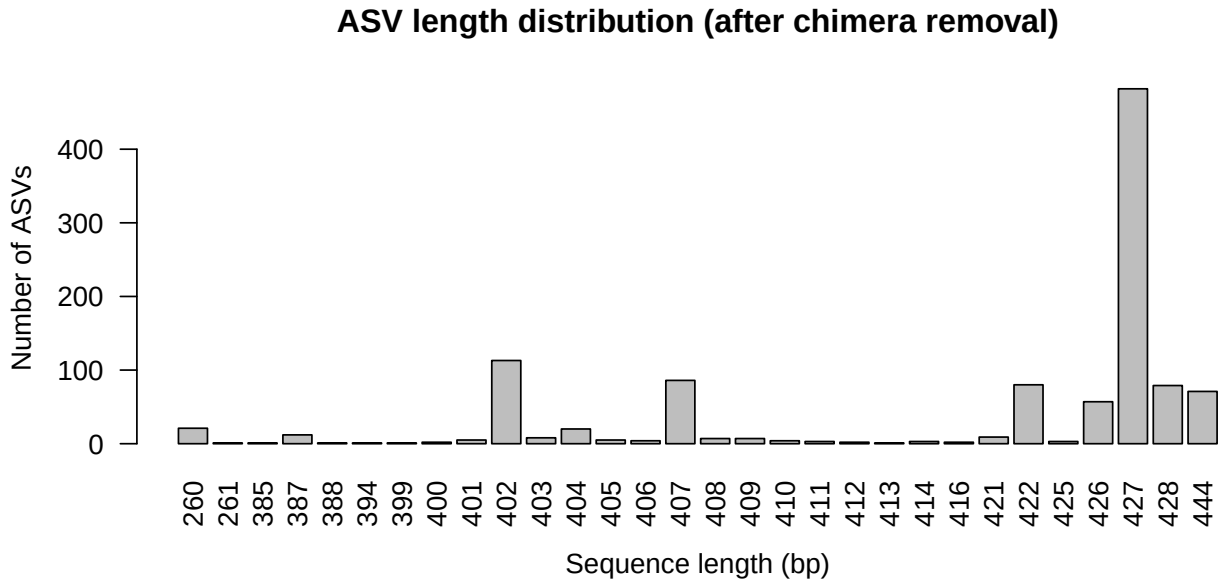
i Interpreting chimera removal

It is **normal** for a large *number* of unique sequences to be flagged as chimeric — often 10–30 % of ASVs. However, chimeric reads usually represent a small proportion of total *reads* (< 10 %), because chimeras are typically rare relative to their parent sequences. If you lose > 20 % of reads here, double-check that primers were fully trimmed.

```
seq_lengths_nc <- table(nchar(getSequences(seqtab.nochim)))
print(seq_lengths_nc)
```

```
260 261 385 387 388 394 399 400 401 402 403 404 405 406 407 408 409 410 411 412
  21   1   1  12   1   1   1   2   5 113   8  20   5   4  86   7   7   4   3   2
413 414 416 421 422 425 426 427 428 444
   1   3   2   9  80   3  57 482  79  71
```

```
barplot(seq_lengths_nc,
        xlab = "Sequence length (bp)",
        ylab = "Number of ASVs",
        main = "ASV length distribution (after chimera removal)",
        las = 2)
```



5.5 Tracking reads through the pipeline

A key quality-control step is to track how many reads survive each stage of the pipeline. Large losses at any single step point to a specific problem.

```
getN <- function(x) sum(getUniques(x))

track <- cbind(
  out,
  denoisedF = sapply(dadaFs, getN),
  denoisedR = sapply(dadaRs, getN),
  merged    = sapply(mergers, getN),
  nonchim   = rowSums(seqtab.nochim)
)
colnames(track) <- c("input", "filtered", "denoisedF", "denoisedR",
  "merged", "nonchim")
rownames(track) <- sample.names

write.csv(track, "dada2_QC.csv")
track
```

	input	filtered	denoisedF	denoisedR	merged	nonchim
Commercial_R1	16778	13226	13183	13200	13113	12499
Commercial_R2	218777	178641	177733	178459	176766	176479
Commercial_R3	14993	12109	12049	12098	11922	11670
Control_extraction	2345	1419	1411	1414	1403	1403
Control_fermentation	42709	34239	34114	34105	33515	33162
Lactobacillus_H24_R1	236996	193289	192552	192803	190009	178597
Lactobacillus_H24_R2	271744	221122	220906	220980	220054	217351
Lactobacillus_H24_R3	185402	150467	150220	150328	149791	148672
Lactobacillus_H48_R1	232087	189922	189536	189665	187248	177255
Lactobacillus_H48_R2	277388	227640	227197	227476	226648	222104
Lactobacillus_H48_R3	257945	210626	210217	210457	209083	205846

Lactobacillus_W1_R1	237242	191040	190633	190702	189362	185573
Lactobacillus_W1_R2	283367	233210	232746	232883	230755	224196
Lactobacillus_W1_R3	135637	107509	107352	107386	106982	105252
Lactobacillus_W2_R1	270194	219755	219374	219573	218558	217417
Lactobacillus_W2_R2	262219	213873	213132	213708	212577	209841
Lactobacillus_W2_R3	280448	230240	229271	229942	227359	218116
Spontaneous_H24_R1	260410	209712	208384	208726	197139	162260
Spontaneous_H24_R2	297828	241630	240341	240755	230290	169513
Spontaneous_H24_R3	258467	209072	208224	208561	202745	166417
Spontaneous_H48_R1	311391	253245	252497	252931	229750	203601
Spontaneous_H48_R2	302367	243119	241992	242408	232963	186846
Spontaneous_H48_R3	295072	238579	237584	238146	230008	193478
Spontaneous_W1_R1	266958	217397	216521	216709	206641	165163
Spontaneous_W1_R2	271112	218668	218010	217981	207723	180948
Spontaneous_W1_R3	295075	239236	238162	238617	230440	187702
Spontaneous_W2_R1	291669	237723	236868	237129	225123	190377
Spontaneous_W2_R2	259494	208982	208097	208521	200762	151955
Spontaneous_W2_R3	314666	256480	255722	255872	248160	210209

5.5.1 Visualise read tracking

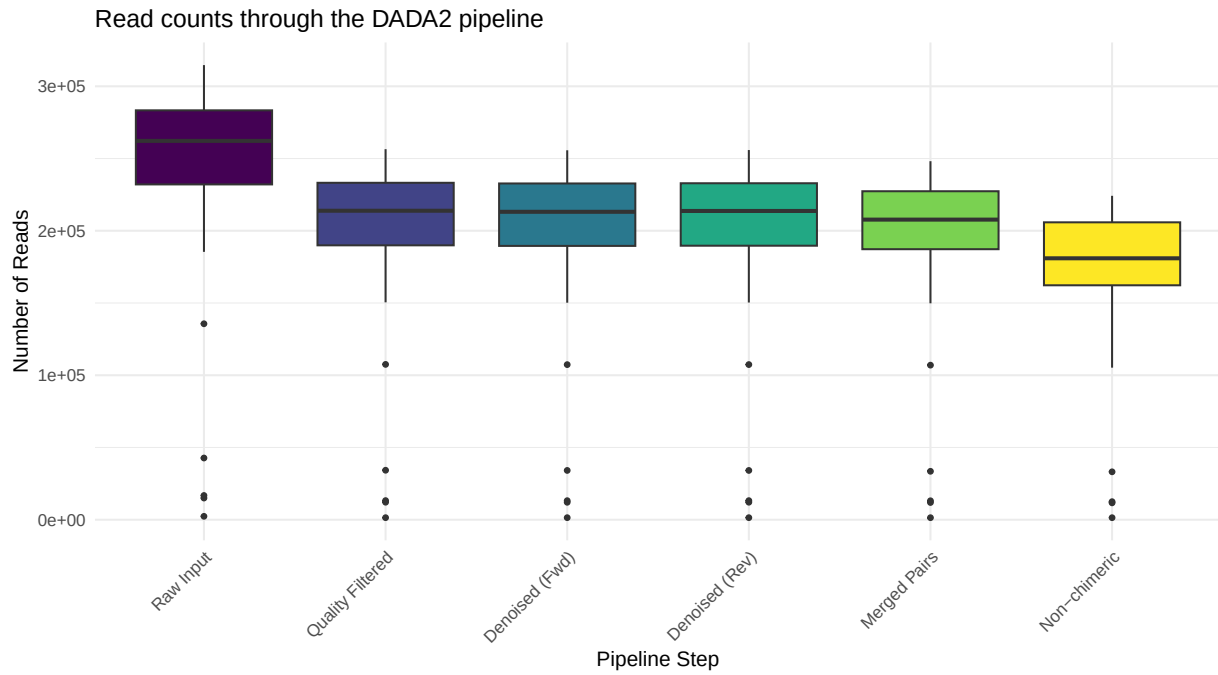
```

step_labels <- c(
  input      = "Raw Input",
  filtered   = "Quality Filtered",
  denoisedF  = "Denoised (Fwd)",
  denoisedR  = "Denoised (Rev)",
  merged     = "Merged Pairs",
  nonchim    = "Non-chimeric"
)

track_long <- as.data.frame(track) %>%
  rownames_to_column("Sample") %>%
  pivot_longer(-Sample, names_to = "Step", values_to = "Reads") %>%
  mutate(Step = factor(Step, levels = names(step_labels)))

ggplot(track_long, aes(x = Step, y = Reads, fill = Step)) +
  geom_boxplot(outlier.size = 1) +
  scale_x_discrete(labels = step_labels) +
  scale_fill_viridis_d() +
  labs(x = "Pipeline Step", y = "Number of Reads",
       title = "Read counts through the DADA2 pipeline") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")

```



What to look for:

- A gradual, smooth decline across steps is expected and healthy
- A sudden large drop at the **filtering** step → truncation lengths may be too aggressive
- A large drop at **merging** → check overlap between reads; check primers are removed
- A large drop at **chimera removal** → investigate whether primers are truly absent

6 Taxonomy Assignment

6.1 Overview

Once we have a table of denoised, chimera-free ASV sequences, we can assign each one a taxonomic identity by comparing it to a reference database. DADA2 uses a **k-mer-based naive Bayesian classifier** (Wang et al. 2007) implemented in `assignTaxonomy()`.

The function assigns taxonomy from Kingdom down to Genus, returning a confidence **bootstrap value** for each level. Assignments with low bootstrap support (< 50 % by default) are reported as NA rather than an unreliable label.

6.2 The SILVA reference database

We use the **SILVA** database (v138), one of the most comprehensive curated 16S/18S rRNA gene databases. We use the formatted training set provided by the DADA2 team:

File	Content
<code>silva_nr_v138_train_set.fa.gz</code>	Kingdom → Genus classifier

Species-level assignment

Species-level assignment requires a second, separate reference file (`silva_v138_species_assignment.fa.gz`). It uses exact matching of the full ASV sequence rather than the Bayesian classifier, and should only be trusted when the amplicon region is discriminating at the species level. We skip `addSpecies()` in this tutorial because the file is not included in the dataset, but it can be added as a straightforward extra step if needed.

6.3 Run taxonomy assignment

```
taxa_file <- file.path(path_precomp, "taxa.rds")

if (!file.exists(taxa_file)) {
  message("Assigning taxonomy (this may take several minutes)...")
  taxa <- assignTaxonomy(seqtab.nochim,
                        path_silva,
                        multithread = TRUE,
                        verbose      = TRUE)
  saveRDS(taxa, taxa_file)
```

```

} else {
  message("Loading pre-computed taxonomy.")
  taxa <- readRDS(taxa_file)
}

```

6.4 Inspect the taxonomy table

```

# Remove sequence row names for readability
taxa.print <- taxa
rownames(taxa.print) <- NULL
head(taxa.print, 10)

```

	Kingdom	Phylum	Class	Order
[1,]	"Bacteria"	"Firmicutes"	"Bacilli"	"Lactobacillales"
[2,]	"Bacteria"	"Firmicutes"	"Bacilli"	"Lactobacillales"
[3,]	"Bacteria"	"Firmicutes"	"Bacilli"	"Lactobacillales"
[4,]	"Bacteria"	"Firmicutes"	"Bacilli"	"Staphylococcales"
[5,]	"Bacteria"	"Firmicutes"	"Bacilli"	"Lactobacillales"
[6,]	"Bacteria"	"Proteobacteria"	"Gammaproteobacteria"	"Enterobacterales"
[7,]	"Bacteria"	"Firmicutes"	"Bacilli"	"Lactobacillales"
[8,]	"Bacteria"	"Proteobacteria"	"Gammaproteobacteria"	"Enterobacterales"
[9,]	"Bacteria"	"Firmicutes"	"Bacilli"	"Lactobacillales"
[10,]	"Bacteria"	"Firmicutes"	"Bacilli"	"Lactobacillales"

	Family	Genus
[1,]	"Lactobacillaceae"	"Lactobacillus"
[2,]	"Lactobacillaceae"	"Lactobacillus"
[3,]	"Streptococcaceae"	"Lactococcus"
[4,]	"Staphylococcaceae"	"Staphylococcus"
[5,]	"Leuconostocaceae"	"Leuconostoc"
[6,]	"Enterobacteriaceae"	"Enterobacter"
[7,]	"Enterococcaceae"	"Enterococcus"
[8,]	"Enterobacteriaceae"	"Enterobacter"
[9,]	"Leuconostocaceae"	"Weissella"
[10,]	"Lactobacillaceae"	"Pediococcus"

The `taxa` object is a matrix with one row per ASV and one column per taxonomic level. NA values indicate the classifier was not sufficiently confident at that level.

6.5 Classification summary

How many ASVs received a classification at each level?

```

taxa_df <- as.data.frame(taxa)
levels <- c("Kingdom", "Phylum", "Class", "Order", "Family", "Genus")

classif <- data.frame(

```



```

Level      = factor(levels, levels = levels),
ASVs_classified = sapply(levels, function(lv) sum(!is.na(taxa_df[[lv]]))),
Total      = nrow(taxa_df)
) %>%
  mutate(Percentage = round(100 * ASVs_classified / Total, 1))

classif

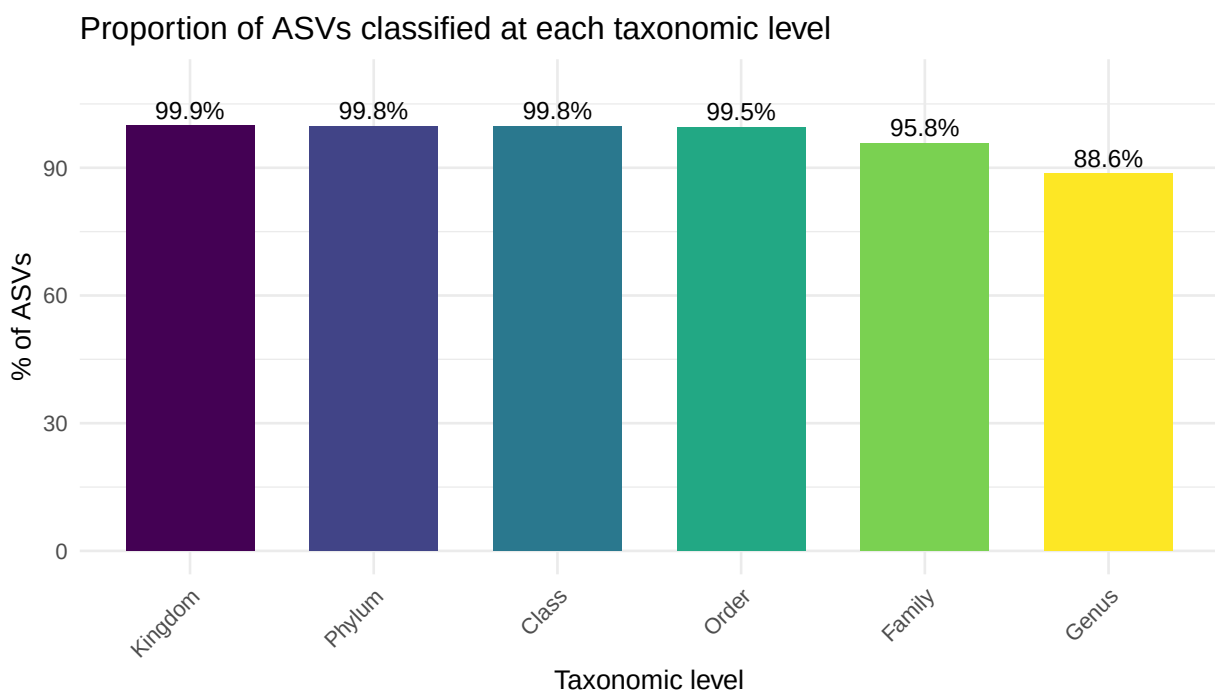
```

	Level	ASVs_classified	Total	Percentage
Kingdom	Kingdom	1090	1091	99.9
Phylum	Phylum	1089	1091	99.8
Class	Class	1089	1091	99.8
Order	Order	1086	1091	99.5
Family	Family	1045	1091	95.8
Genus	Genus	967	1091	88.6

```

ggplot(classif, aes(x = Level, y = Percentage, fill = Level)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = paste0(Percentage, "%")),
            vjust = -0.4, size = 3.5) +
  scale_fill_viridis_d() +
  labs(title = "Proportion of ASVs classified at each taxonomic level",
       x = "Taxonomic level",
       y = "% of ASVs") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none") +
  ylim(0, 110)

```



 Interpreting classification depth

- Near 100 % classification at **Kingdom** and **Phylum** is expected for a clean 16S dataset
- Classification depth drops progressively at lower levels (Family → Genus) — this is normal and reflects the limits of 16S resolution
- Very low Phylum-level classification (< 80 %) may indicate: non-bacterial sequences, contamination, or a mismatch between the amplified region and the database

7 Phyloseq Integration and Export

7.1 What is phyloseq?

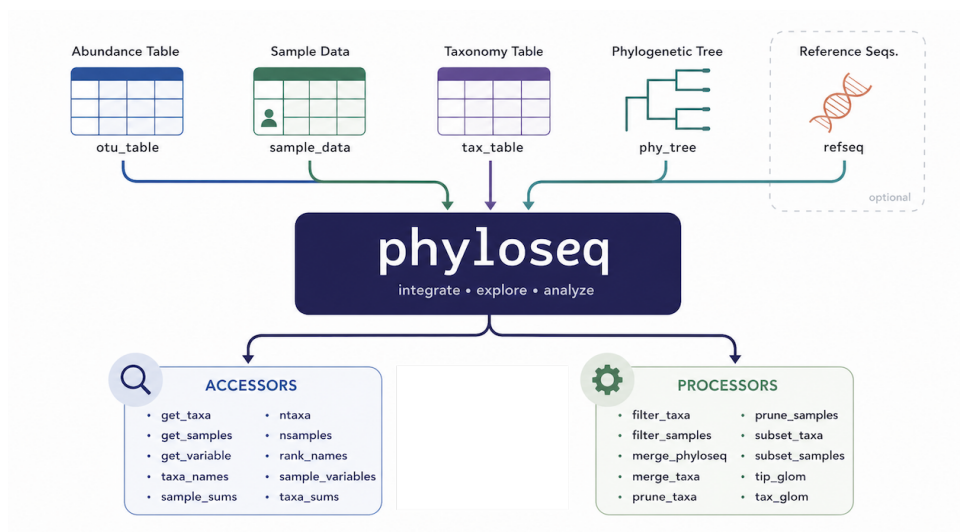


Figure 7.1: Phyloseq

phyloseq is an R package that combines three data tables into a single, consistent object:

Component	Function	Description
OTU table	<code>otu_table()</code>	ASV × sample abundance matrix
Taxonomy table	<code>tax_table()</code>	ASV × taxonomic level annotations
Sample data	<code>sample_data()</code>	Sample-level metadata
Reference sequences	<code>refseq()</code>	The actual DNA sequences (optional)

Keeping everything in one object prevents accidental misalignment between tables and gives access to a rich set of analysis and visualisation functions.

7.2 Prepare metadata

We derive metadata directly from the sample names, which encode the experimental design (Group_Timepoint_Replicate).

```
# Parse sample names into experimental variables
metadata <- data.frame(
  SampleID = sample.names,
  row.names = sample.names,
  stringsAsFactors = FALSE
) %>%
  mutate(
    Group = case_when(
      startsWith(SampleID, "Lactobacillus") ~ "Lactobacillus",
      startsWith(SampleID, "Spontaneous") ~ "Spontaneous",
      startsWith(SampleID, "Commercial") ~ "Commercial",
      startsWith(SampleID, "Control") ~ "Control",
      TRUE ~ "Unknown"
    ),
    Timepoint = case_when(
      grepl("_H24_", SampleID) ~ "H24",
      grepl("_H48_", SampleID) ~ "H48",
      grepl("_W1_", SampleID) ~ "W1",
      grepl("_W2_", SampleID) ~ "W2",
      TRUE ~ NA_character_
    ),
    Replicate = as.integer(sub(".*_R(\\d+)$", "\\1", SampleID)),
    IsControl = Group == "Control"
  )

# Verify that metadata rows match the seqtab rows
stopifnot(all(rownames(seqtab.nochim) %in% rownames(metadata)))

metadata
```

	SampleID	Group	Timepoint	Replicate
Commercial_R1	Commercial_R1	Commercial	<NA>	1
Commercial_R2	Commercial_R2	Commercial	<NA>	2
Commercial_R3	Commercial_R3	Commercial	<NA>	3
Control_extraction	Control_extraction	Control	<NA>	NA
Control_fermentation	Control_fermentation	Control	<NA>	NA
Lactobacillus_H24_R1	Lactobacillus_H24_R1	Lactobacillus	H24	1
Lactobacillus_H24_R2	Lactobacillus_H24_R2	Lactobacillus	H24	2
Lactobacillus_H24_R3	Lactobacillus_H24_R3	Lactobacillus	H24	3
Lactobacillus_H48_R1	Lactobacillus_H48_R1	Lactobacillus	H48	1
Lactobacillus_H48_R2	Lactobacillus_H48_R2	Lactobacillus	H48	2
Lactobacillus_H48_R3	Lactobacillus_H48_R3	Lactobacillus	H48	3
Lactobacillus_W1_R1	Lactobacillus_W1_R1	Lactobacillus	W1	1
Lactobacillus_W1_R2	Lactobacillus_W1_R2	Lactobacillus	W1	2
Lactobacillus_W1_R3	Lactobacillus_W1_R3	Lactobacillus	W1	3
Lactobacillus_W2_R1	Lactobacillus_W2_R1	Lactobacillus	W2	1
Lactobacillus_W2_R2	Lactobacillus_W2_R2	Lactobacillus	W2	2
Lactobacillus_W2_R3	Lactobacillus_W2_R3	Lactobacillus	W2	3
Spontaneous_H24_R1	Spontaneous_H24_R1	Spontaneous	H24	1
Spontaneous_H24_R2	Spontaneous_H24_R2	Spontaneous	H24	2
Spontaneous_H24_R3	Spontaneous_H24_R3	Spontaneous	H24	3

Spontaneous_H48_R1	Spontaneous_H48_R1	Spontaneous	H48	1
Spontaneous_H48_R2	Spontaneous_H48_R2	Spontaneous	H48	2
Spontaneous_H48_R3	Spontaneous_H48_R3	Spontaneous	H48	3
Spontaneous_W1_R1	Spontaneous_W1_R1	Spontaneous	W1	1
Spontaneous_W1_R2	Spontaneous_W1_R2	Spontaneous	W1	2
Spontaneous_W1_R3	Spontaneous_W1_R3	Spontaneous	W1	3
Spontaneous_W2_R1	Spontaneous_W2_R1	Spontaneous	W2	1
Spontaneous_W2_R2	Spontaneous_W2_R2	Spontaneous	W2	2
Spontaneous_W2_R3	Spontaneous_W2_R3	Spontaneous	W2	3
IsControl				
Commercial_R1	FALSE			
Commercial_R2	FALSE			
Commercial_R3	FALSE			
Control_extraction	TRUE			
Control_fermentation	TRUE			
Lactobacillus_H24_R1	FALSE			
Lactobacillus_H24_R2	FALSE			
Lactobacillus_H24_R3	FALSE			
Lactobacillus_H48_R1	FALSE			
Lactobacillus_H48_R2	FALSE			
Lactobacillus_H48_R3	FALSE			
Lactobacillus_W1_R1	FALSE			
Lactobacillus_W1_R2	FALSE			
Lactobacillus_W1_R3	FALSE			
Lactobacillus_W2_R1	FALSE			
Lactobacillus_W2_R2	FALSE			
Lactobacillus_W2_R3	FALSE			
Spontaneous_H24_R1	FALSE			
Spontaneous_H24_R2	FALSE			
Spontaneous_H24_R3	FALSE			
Spontaneous_H48_R1	FALSE			
Spontaneous_H48_R2	FALSE			
Spontaneous_H48_R3	FALSE			
Spontaneous_W1_R1	FALSE			
Spontaneous_W1_R2	FALSE			
Spontaneous_W1_R3	FALSE			
Spontaneous_W2_R1	FALSE			
Spontaneous_W2_R2	FALSE			
Spontaneous_W2_R3	FALSE			

7.3 Build the phyloseq object

```
# Ensure tables are in the same order
metadata_matched <- metadata[rownames(seqtab.nochim), ]
seqtab_matched   <- seqtab.nochim[rownames(metadata_matched), ]

stopifnot(all(rownames(metadata_matched) == rownames(seqtab_matched)))

otu <- otu_table(t(seqtab_matched), taxa_are_rows = TRUE)
```

```
tax <- tax_table(taxa)
sd <- sample_data(metadata_matched)

ps <- phyloseq(otu, sd, tax)

# Store the actual DNA sequences in the phyloseq object
dna <- Biostrings::DNASTringSet(taxa_names(ps))
names(dna) <- taxa_names(ps)
ps <- merge_phyloseq(ps, dna)

# Rename ASVs to simple identifiers (ASV1, ASV2, ...)
taxa_names(ps) <- paste0("ASV", seq(ntaxa(ps)))

ps
```

```
phyloseq-class experiment-level object
otu_table() OTU Table: [ 1091 taxa and 29 samples ]
sample_data() Sample Data: [ 29 samples by 5 sample variables ]
tax_table() Taxonomy Table: [ 1091 taxa by 6 taxonomic ranks ]
refseq() DNASTringSet: [ 1091 reference sequences ]
```

7.4 Filter unwanted taxa

16S amplification can co-amplify **mitochondrial** and **chloroplast** 16S-like sequences from plant and eukaryotic cells. These are not bacteria and must be removed before downstream analyses.

```
ps_filtered <- ps %>%
  subset_taxa(
    (is.na(Family) | !grepl("mitochondria", Family, ignore.case = TRUE)) &
    (is.na(Order) | !grepl("mitochondria", Order, ignore.case = TRUE)) &
    (is.na(Genus) | !grepl("mitochondria", Genus, ignore.case = TRUE)) &
    (is.na(Family) | !grepl("chloroplast", Family, ignore.case = TRUE)) &
    (is.na(Order) | !grepl("chloroplast", Order, ignore.case = TRUE)) &
    (is.na(Order) | Order != "Chloroplastida")
  )

cat("ASVs removed (mitochondria / chloroplasts):",
    ntaxa(ps) - ntaxa(ps_filtered), "\n")
```

```
ASVs removed (mitochondria / chloroplasts): 26
```

7.4.1 Remove ASVs unclassified at Family level

For most downstream community analyses we require at least Family-level classification. ASVs that could not be classified even at this level are removed.

```
ps_final <- ps_filtered %>%
  subset_taxa(!is.na(Family) & Family != "")

cat("ASVs removed (unclassified at Family):",
    ntaxa(ps_filtered) - ntaxa(ps_final), "\n")
```

ASVs removed (unclassified at Family): 27

```
cat("Final ASV count:", ntaxa(ps_final), "\n")
```

Final ASV count: 1038

```
cat("Total reads in final dataset:", sum(sample_sums(ps_final)), "\n")
```

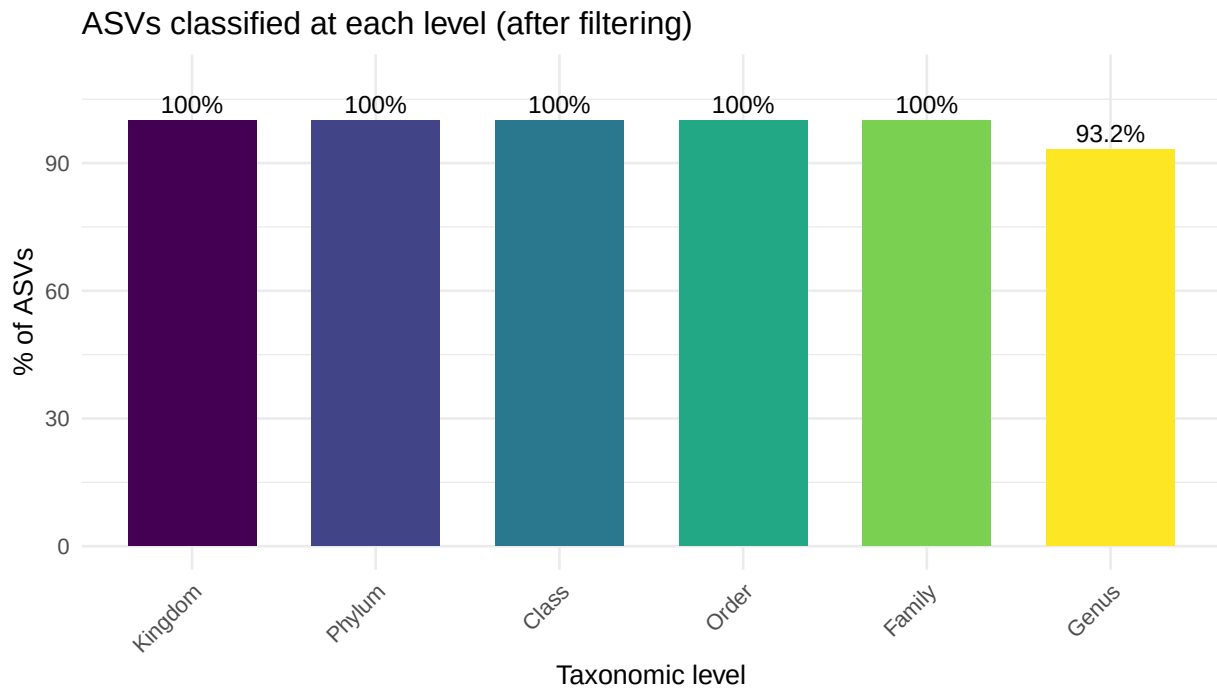
Total reads in final dataset: 4674036

7.5 Taxonomic composition overview

```
tax_df <- as.data.frame(tax_table(ps_final))
levels <- c("Kingdom", "Phylum", "Class", "Order", "Family", "Genus")
total <- ntaxa(ps_final)

tax_classif <- data.frame(
  Level = factor(levels, levels = levels),
  Pct = sapply(levels, function(lv) {
    round(100 * sum(!is.na(tax_df[[lv]]) & tax_df[[lv]] != "") / total, 1)
  })
)

ggplot(tax_classif, aes(x = Level, y = Pct, fill = Level)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = paste0(Pct, "%")), vjust = -0.4, size = 3.5) +
  scale_fill_viridis_d() +
  labs(title = "ASVs classified at each level (after filtering)",
       x = "Taxonomic level", y = "% of ASVs") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none") +
  ylim(0, 110)
```



7.6 Filtering summary

```
cat("\n=== FILTERING SUMMARY ===\n")
```

```
=== FILTERING SUMMARY ===
```

```
cat("After denoising and chimera removal:      ", ntaxa(ps), "ASVs\n")
```

```
After denoising and chimera removal:      1091 ASVs
```

```
cat("After removing mitochondria/chloroplasts: ", ntaxa(ps_filtered), "ASVs\n")
```

```
After removing mitochondria/chloroplasts: 1065 ASVs
```

```
cat("After removing unclassified at Family:     ", ntaxa(ps_final), "ASVs\n")
```

```
After removing unclassified at Family:     1038 ASVs
```

```
cat("Samples in final dataset:                   ", nsamples(ps_final), "\n")
```

```
Samples in final dataset:                   29
```



```
cat("Total reads in final dataset:           ", sum(sample_sums(ps_final)), "\n")
```

```
Total reads in final dataset:           4674036
```

7.7 Export outputs

```
# Phyloseq object (for downstream analysis in R)
saveRDS(ps_final, "phyloseq_object.rds")

# ASV ID → sequence mapping
asv_sequences <- data.frame(
  ASV_ID    = taxa_names(ps_final),
  Sequence  = as.character(refseq(ps_final)),
  stringsAsFactors = FALSE
)
write.csv(asv_sequences, "asv_sequences_map.csv", row.names = FALSE)

# ASV taxonomy + abundance table
asv_counts    <- as.data.frame(otu_table(ps_final))
tax_table_df  <- as.data.frame(tax_table(ps_final))
write.csv(cbind(tax_table_df, asv_counts),
          "asv_tax_table.csv", row.names = TRUE)

# Sample metadata + per-sample read counts per ASV
sample_abundances <- as.data.frame(t(otu_table(ps_final)))
sample_metadata   <- as.data.frame(sample_data(ps_final))
write.csv(cbind(sample_metadata, sample_abundances),
          "samples_with_abundances.csv", row.names = TRUE)

cat("Outputs saved:\n")
```

```
Outputs saved:
```

```
cat("  phyloseq_object.rds\n")
```

```
  phyloseq_object.rds
```

```
cat("  asv_sequences_map.csv\n")
```

```
  asv_sequences_map.csv
```

```
cat("  asv_tax_table.csv\n")
```

```
  asv_tax_table.csv
```

```
cat("  samples_with_abundances.csv\n")
```

```
samples_with_abundances.csv
```

7.8 What's next?

The `phyloseq_object.rds` file is the starting point for all downstream analyses:

- **Alpha diversity** — species richness, Shannon entropy, Faith's PD
- **Beta diversity** — Bray-Curtis or UniFrac distances, ordination (PCoA, NMDS)
- **Differential abundance** — DESeq2, ANCOM-BC, MaAsLin2
- **Taxonomic bar charts** — relative abundance at Phylum or Family level
- **Network analysis** — co-occurrence networks using SpiecEasi or SPRING

These topics are covered in the downstream analysis tutorial.